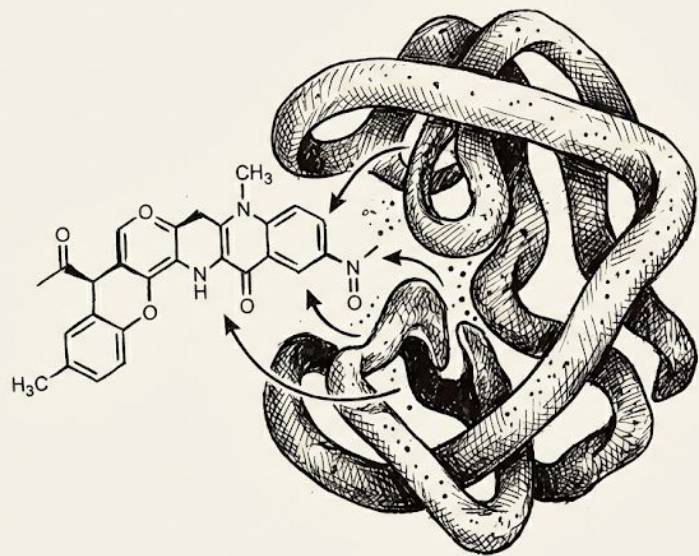




# DeepGLSTM

## Graph Based Methods for Predicting Drug-Target Binding Affinity

**Kovvada Dhyana Vardhan (21CS30030)**

Research for future work, Making Presentation

**Tanishq Prasad (21CS30054)**

Writing the Report, Executing code, Working on Improvements, Refactoring PPT

**Ayush Sunil Munot (21ME3FP55)**

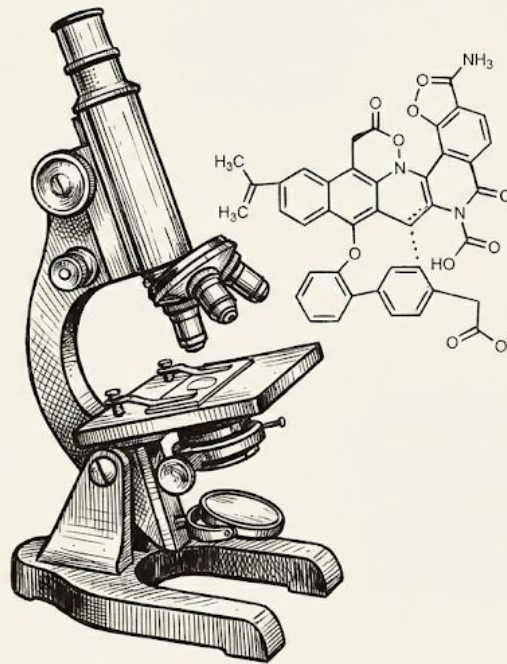
Codebase Refactoring, Reproduction of Results, Ablation Studies, Refactoring PPT

# Overview and Agenda

- Solving the **drug-target binding** prediction problem.
- Reviewing previous solutions and their limitations.
- Introduction to the novel **DeepGLSTM** model architecture.
- Experimental results, reproduction, and quantitative analysis.
- Discussion of potential improvements and future work.

# Problem Statement

- Predicting if a drug binds with a protein for computational drug discovery
- Drug repurposing on new target proteins
- Using **Machine Learning and Deep Learning concepts** to learn drug-target interaction.



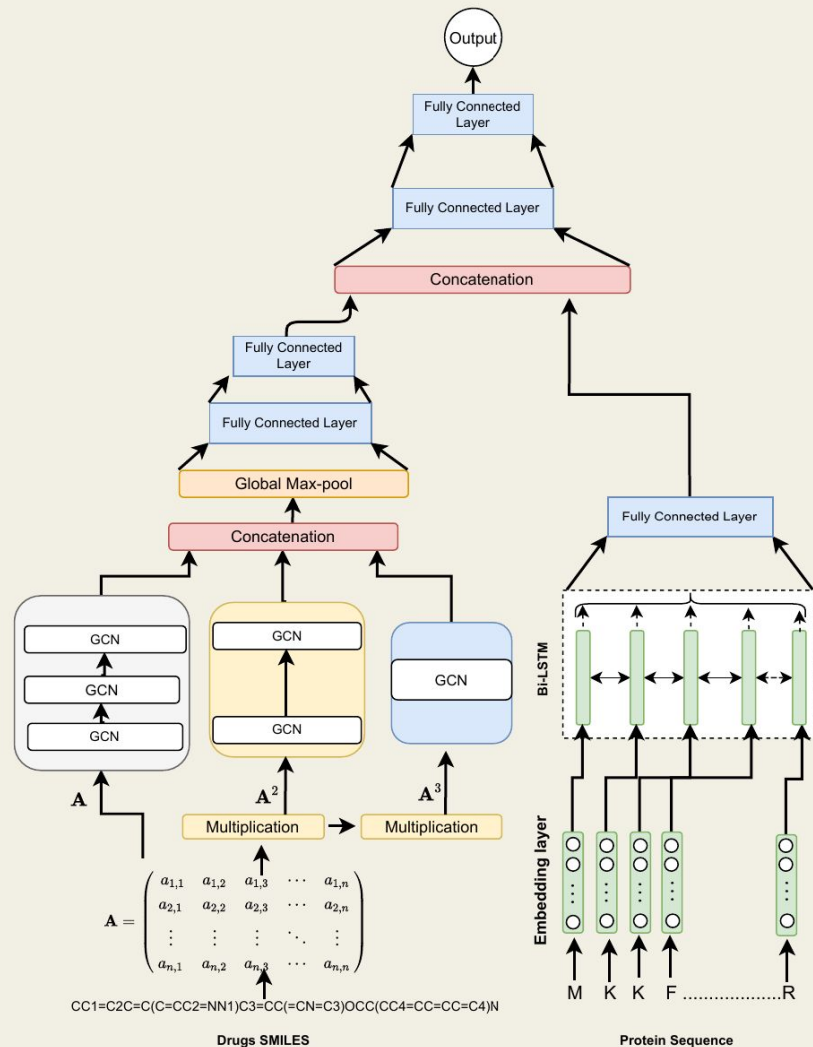
# Limitations of previous solutions

- Traditional methods are costly and time-consuming.
- Examining interaction between drugs and proteins results in high throughput.
- Statistical and simple machine learning models **e.g. SimBoost, Kronecker, etc.** highly rely on feature engineering.
- Earlier deep learning models (**DTA models like DeepDTA, WideDTA**) use string representation, losing topological information.
- **GNN-based** did not work properly capture topological because it provides simple adjacency representation as input and used 1-D CNN for protein sequencing data.

# DeepGLSTM

- Better accuracy prediction score using two main functional modules.
- **SMILES** notation is used to represent the drug compound molecules as a graph, using **RDKit**.
- The model uses **GCN blocks** to capture topological information from drugs with adjacency matrices upto **two hop distance**, resulting in **graph embedding**.
- **Bi-LSTM** is used to capture sequential information from protein sequences converting them into **protein embeddings**.
- Followed by **concatenation of the embeddings** which is then passed through fully connected layers.

# Model architecture of DeepGLSTM



# Datasets

| Dataset       | # of drugs<br>(compounds) | # of targets<br>(proteins) | Total number of<br>drug-target pairs |
|---------------|---------------------------|----------------------------|--------------------------------------|
| Davis(kd)     | 68                        | 442                        | 30056                                |
| KIBA          | 2111                      | 229                        | 118254                               |
| DTC(pki)      | 5983                      | 118                        | 67894                                |
| Metz(pki)     | 1471                      | 170                        | 35307                                |
| ToxCast(AC50) | 7657                      | 328                        | 342869                               |
| STITCH        | 724471                    | 15258                      | 1244420                              |

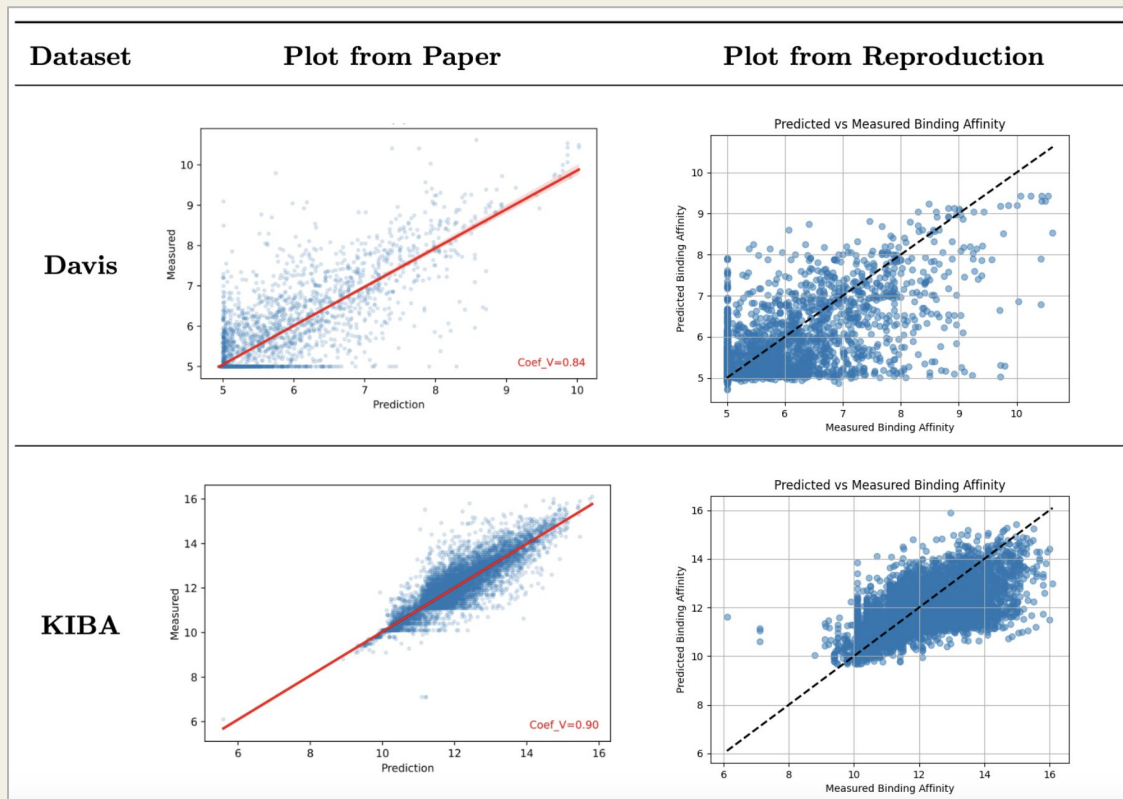
- **Subset Utilization:** Utilized a **~30% random subset** of the Davis, KIBA, and Metz datasets to make training feasible within limited compute hours.
- **Strategic Omissions:** Excluded the largest datasets (STITCH, ToxCast, DTC) as their massive size exceeded available memory resources

# Experimental Results

| Dataset | MSE   |                    | CI    |                    | r <sup>2</sup> <sub>m</sub> |                    |
|---------|-------|--------------------|-------|--------------------|-----------------------------|--------------------|
|         | Paper | Reproduced Results | Paper | Reproduced Results | Paper                       | Reproduced Results |
| Davis   | 0.232 | 0.387              | 0.895 | 0.832              | 0.680                       | 0.479              |
| KIBA    | 0.133 | 0.382              | 0.897 | 0.768              | 0.792                       | 0.412              |
| Metz    | 0.294 | 0.41               | 0.810 | 0.767              | 0.640                       | 0.516              |

- **Robust Ranking Capability:** Achieved a strong **Concordance Index (CI) of 0.832** on Davis, proving the model correctly distinguishes high vs. low affinity pairs despite limited training data.
- **Expected Performance Gap: MSE increased** (0.387 vs. 0.232) and r<sub>m</sub><sup>2</sup> dropped as a direct result of training on a **30% data subset**.
- **Successful Convergence:** The model effectively learned predictive patterns across all metrics (MSE, CI, r<sub>m</sub><sup>2</sup>) within just **400 epochs**, demonstrating architectural robustness.

# Qualitative Analysis



- **Correlation Confirmed:** Reproduced plots track the diagonal ( $y=x$ ) effectively, validating that the model learned the fundamental binding physics despite constraints .
- **Variance vs. Data (Davis):** The reproduction shows a **wider prediction spread** (visual noise) compared to the paper's tight clustering, directly resulting from training on only **30% of the data** .
- **Saturation Effect (KIBA):** High-density region shows model slightly under-predicts extreme high-affinity values (top-right), side effect of reducing training from 1000 to 400 epochs.

# Ablation Analysis

Effectiveness of different components of DeepGLSTM (using Davis Dataset (pKd)

| Experiment No. | Name of the Models                                                                                                       | MSE (Paper)  | MSE(Reproduced Results) |
|----------------|--------------------------------------------------------------------------------------------------------------------------|--------------|-------------------------|
| R1             | 1st block→ 1st block + 2nd block (# of layers 2)                                                                         | 0.237        | 0.438                   |
| R2             | <b>1st block + 2nd block→ DeepGLSTM</b>                                                                                  | <b>0.232</b> | <b>0.429</b>            |
| R3             | DeepGLSTM → 1st block (# of layers 4) + 2nd block (# of layers 3) + 3rd block (# of layers 2) + 4th block (# of layer 1) | 0.238        | 0.445                   |

- **Trend Validation:** The reproduced results perfectly mirror the original paper pattern ( $R2 < R1 < R3$ ), confirming that the specific 3-block architecture is optimal.
- **The "Sweet Spot": DeepGLSTM (R2)** achieved the lowest error (**0.429**), outperforming the shallower R1 model (**0.438**).
- **Diminishing Returns:** Increasing depth further (R3) degraded performance (**0.445**), indicating that adding a 4th block causes overfitting rather than improving feature extraction.

# Ablation Analysis

Effectiveness of using the power graph (using Davis Dataset (pKd))

| Models        | Input to GCN block                 | MSE (Paper) | MSE (Reproduced Results) |
|---------------|------------------------------------|-------------|--------------------------|
| 1st block GCN | A                                  | 0.239       | 0.442                    |
| 2nd block GCN | A <sup>2</sup>                     | 0.245       | 0.443                    |
| 3rd block GCN | A <sup>3</sup>                     | 0.244       | 0.467                    |
| DeepGLSTM     | A, A <sup>2</sup> , A <sup>3</sup> | 0.232       | 0.459                    |

- **Individual Strength:** Single blocks using **A** and **A<sup>2</sup>** achieved the lowest errors (MSE ~0.44), confirming they capture the most critical structural information.
- **Unexpected Deviation:** Unlike the paper, the **Combined DeepGLSTM (0.459)** did *not* outperform the single blocks in our reproduction.
- **Data Sensitivity:** The complex multi-hop correlations (especially **A<sup>3</sup>**) failed to improve performance under limited training (100 epochs), suggesting they require the full dataset to be effective.

# Improvements and Future Work

- Consider using Alphafold to generate protein graph structures.
- Introduce cross attention instead of concatenation of drug and protein embeddings for better interpretability.
- Extract subgraphs from the drug structure depending on their importance in binding affinity
- Replacing BiLSTM with a pre trained encoder decoder model , e.g. ESM , helps in long range dependencies
- Learn drug structural patterns and invariances for generalization of the model and faster convergence

# Improvements and Future Work

- Add Graph attention networks or graph isomorphism networks to GCN layers
- Multiscale protein modeling, k-mer embeddings and parallel CNN + BiLSTM
- Test the model with different types of evaluation metrics other than MSE and CI
- Integrate raw 3D coordinate data directly using  $E(n)$ -equivariant GNNs.

**Questions?**

**THANK YOU**